Nitheesha Moturu

GitHub — LinkedIn — LeetCode
nitheeshamoturu@gmail.com — +91 8341325899

Education

Shiv Nadar University Chennai Aug 2024 – Aug 2028
B.Tech in Artificial Intelligence and Data Science
CGPA: 8.686

Experience

Deloitte Australia – Data Analytics Job Simulation 2026

- Analyzed employee compensation data across multiple factories and job roles to identify equality patterns and potential pay disparities.
- Created data visualizations and bar charts to compare equality scores and highlight areas requiring further investigation.
- Developed an interactive Tableau dashboard that enabled users to explore compensation patterns across factories and job roles.
- **Tools:** Tableau, Microsoft Excel
- **Proof of Work:** Completion certificate

Projects

GigShield

- Developed a parametric income protection platform for Q-Commerce delivery partners to provide automated compensation during disruptions such as heavy rainfall, waterlogging, bandh, and curfew.
- Automated the claim evaluation and payout workflow based on predefined disruption conditions, reducing the need for manual claim processing.
- Designed multiple insurance plans with different coverage levels, enabling delivery partners to select protection based on their requirements.
- **Technologies:** React.js, Node.js, Express.js, Python, Machine Learning, PostgreSQL, Render.
- **Live:** gig-shield-main.vercel.app

Retrieval-Augmented Generation (RAG) Document QA System

- Engineered an advanced RAG pipeline with hybrid retrieval (Dense + BM25), Cross-Encoder re-ranking, query rewriting, conversational memory, corrective retrieval, context sufficiency checks, hallucination guardrails, and quantization, improving answer accuracy by 30% and reliability by 40%.
- Implemented performance tracking and source-level citations with Gemini-based generation; containerized and deployed the Streamlit application using Docker, Docker Hub, and Render.
- **Technologies:** Python, RAG, LLMs, Gemini API, Sentence Transformers, FAISS, BM25, Cross-Encoder, Embeddings, Hybrid Search, Re-ranking, Query Rewriting, Quantization, Guardrails, Streamlit, Docker, Render, Git/GitHub.
- **Live:** https://rag-project-7u4g.onrender.com
- **GitHub:** https://github.com/Nitheesha-1025/RAG

AgriFair

- Built an agriculture marketplace with an AI-based crop price prediction system using historical market data, helping users make data-driven pricing decisions with 30% improved price estimation.
- Integrated real-time price predictions using crop, location, market, variety, grade, and seasonal data, with PostgreSQL/Supabase storage for faster access to previous predictions.
- **Technologies:** Python, Scikit-learn, Random Forest, FastAPI, REST API, PostgreSQL, Supabase, SQL, HTML, CSS, JavaScript, Git/GitHub.
- **Live:** agrifair-16e4.onrender.com

Technical Skills

- **Languages:** C++, Java, Python, C, JavaScript, SQL
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, REST APIs
- **AI/ML:** Machine Learning, Natural Language Processing, Deep Learning, Large Language Models, Retrieval-Augmented Generation
- **Libraries/Frameworks:** Scikit-learn, Sentence Transformers, FAISS, FastAPI, React.js, SpringBoot,SpringAI, Streamlit.
- **Databases:** MySQL, PostgreSQL, Supabase
- **Tools:** Git, GitHub, VS Code

Achievements & Honors

- NSS Certification
- NPTEL Certification on "Introduction to LLMs" – Top 1% marks secured [Certificate]
- Guidewire Hackathon – Qualified for the semi-finals, ranking among the Top 20 teams on the leaderboard

Interests

- Building Scalable Applications
- Backend Development
- Artificial Intelligence